

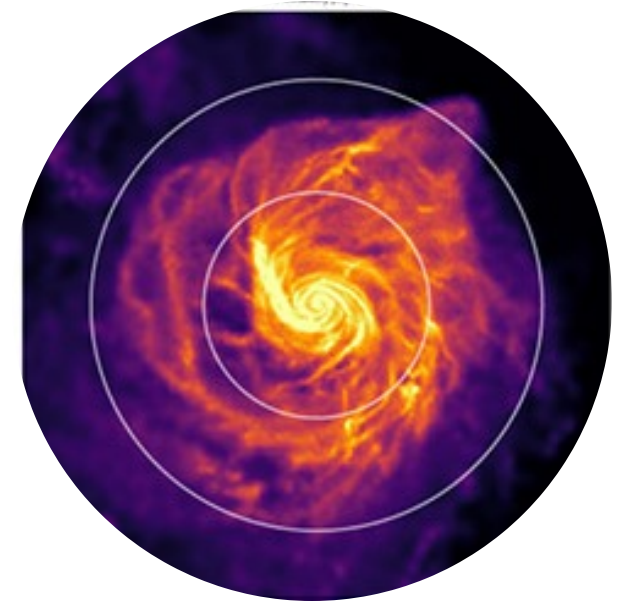
Morphological Characterization of Spiral Arms in IllustrisTNG50 Disk Galaxies and Their Relation to Dark Matter Halo Properties

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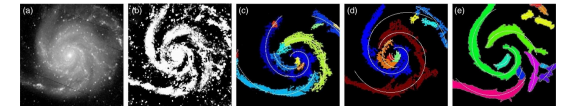
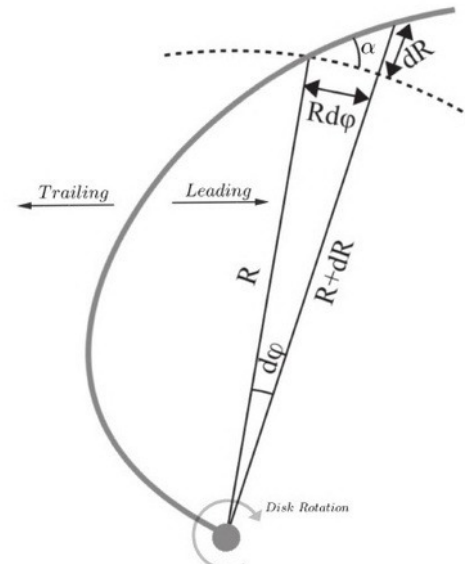
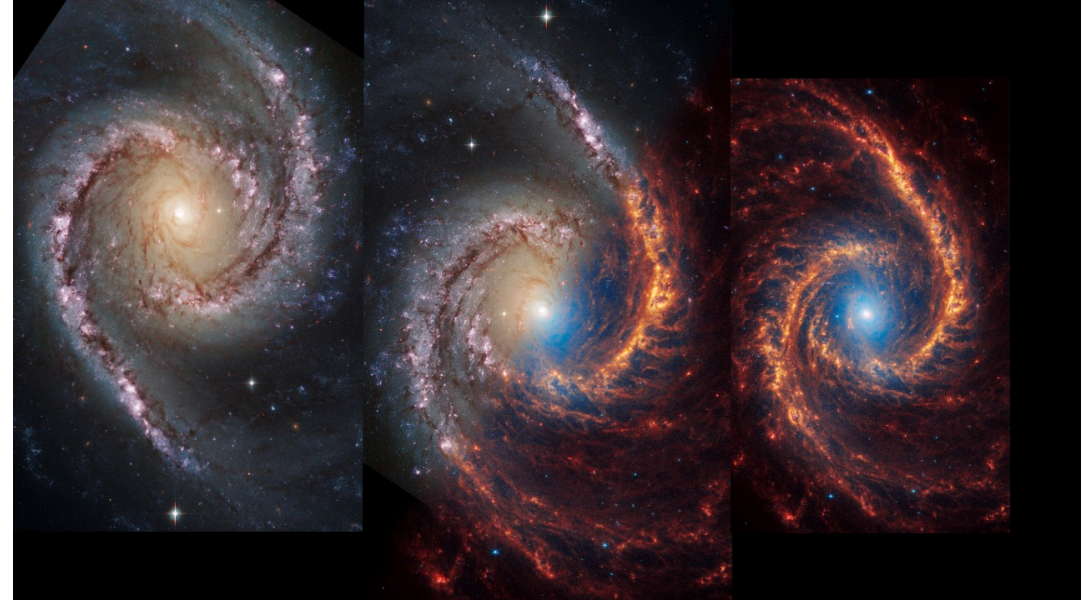
Spiral Arms in Disk Galaxies

Spiral arms appear in several visual styles:

- **~10%** grand-design spirals with two well-defined arms
- **~60%** multi-armed spirals
- **~30%** flocculent spirals without clear global arms

But what are they? How do they form? How do they move?

Even defining one “arm” consistently is difficult.



How do properties of a galaxy's host dark matter halo relate to the morphology of its spiral arms?

Searching for disk–halo correlations can provide discriminating tests among density-wave, swing-amplification, and interaction scenarios.

What Sample Did We Use?

- IllustrisTNG50 at $z = 0$
- 2.7 TB snapshot on disk
- Gas mass $> 10^{9.6} M_{\odot}$ for preselection
- 377 subhalos initially selected
- **309 subhalos passed the disk filters**

Gas cells

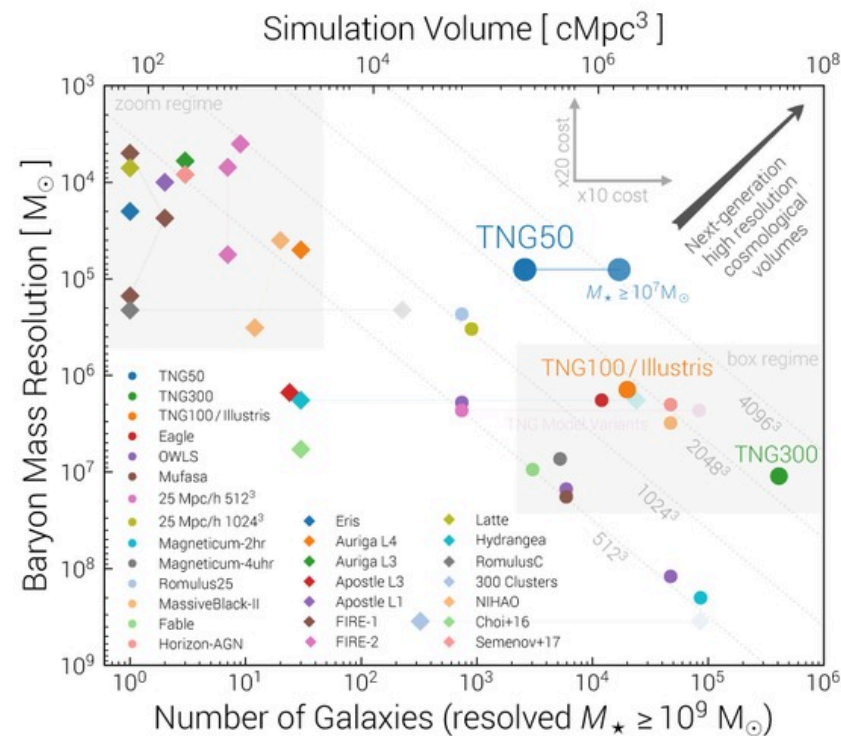
position, velocity,
density, internal
energy

Subhalos

gas, stellar, and dark
matter mass

Structure

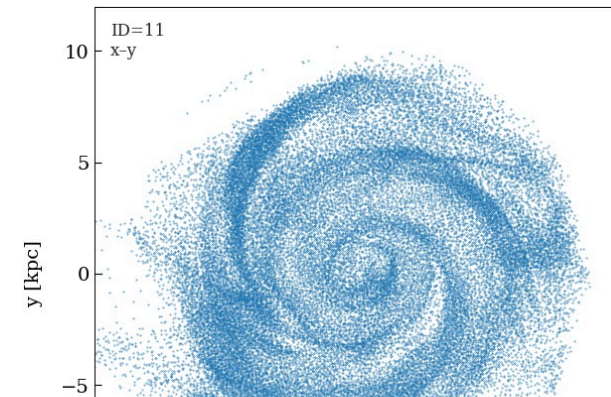
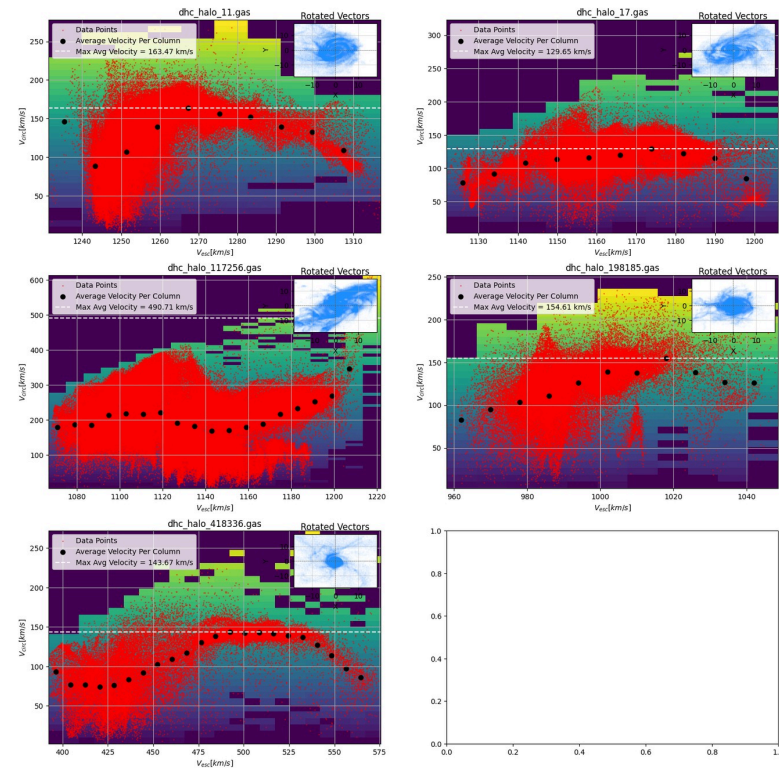
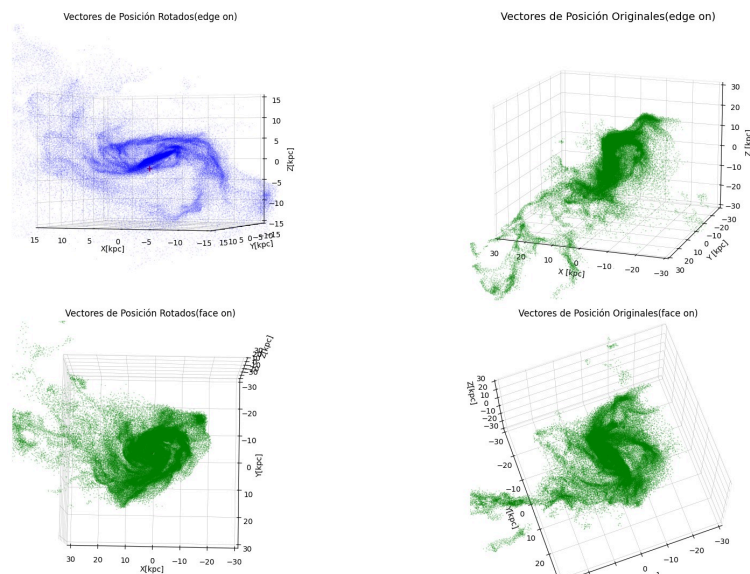
radius, velocity scale,
concentration



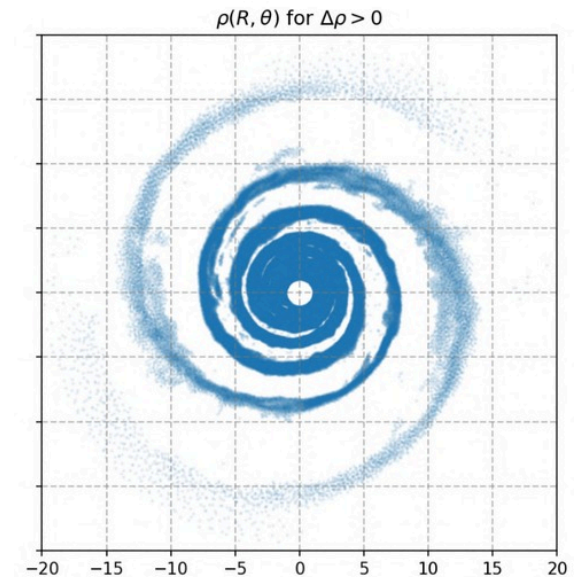
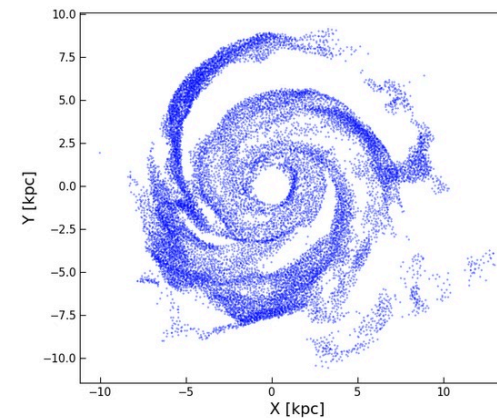
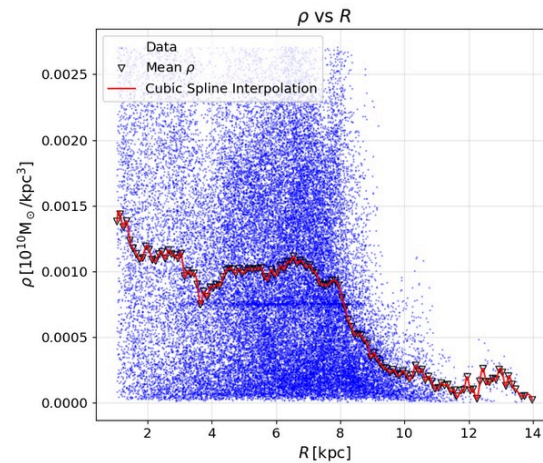
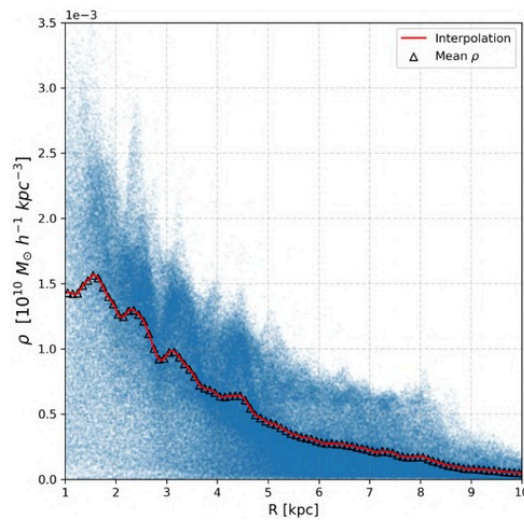
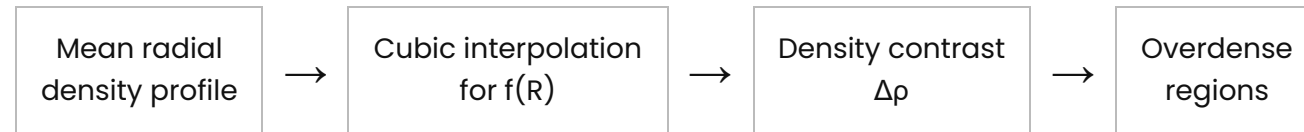
IllustrisTNG simulation volume and baryonic mass resolution. Pillepich et al. (2019).

Selecting and Filtering Disk Galaxies

- Visual inspection of the gas disk
- Spatial density clustering with DBSCAN
- Internal-energy selection to isolate disk gas
- Orbital-circularity selection
- Orientation and face-on deprojection
- **309 subhalos passed all filters**



Measuring Morphological Properties

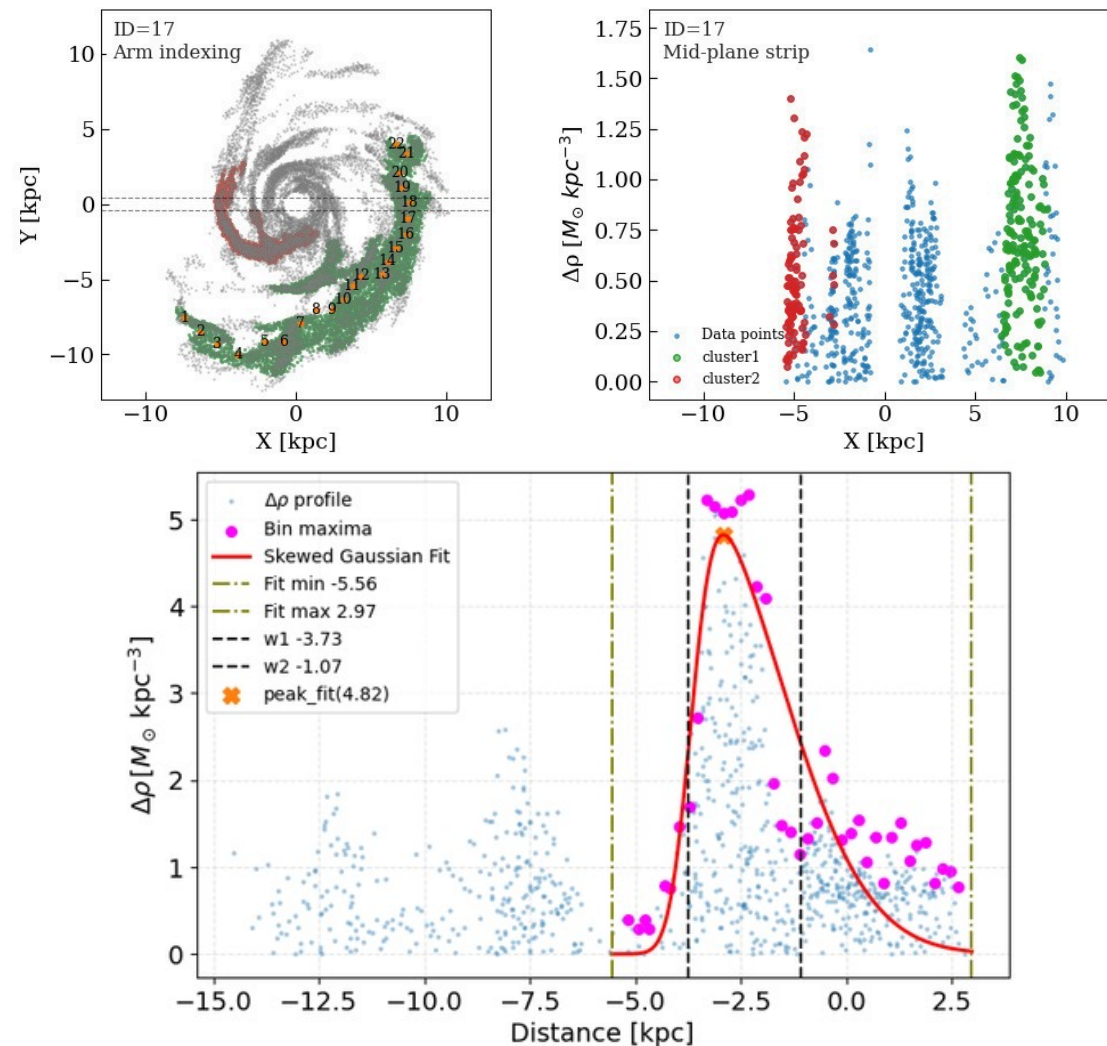


$\Delta\rho < 0$: background particles. $\Delta\rho > 0$: candidate overdense structures.

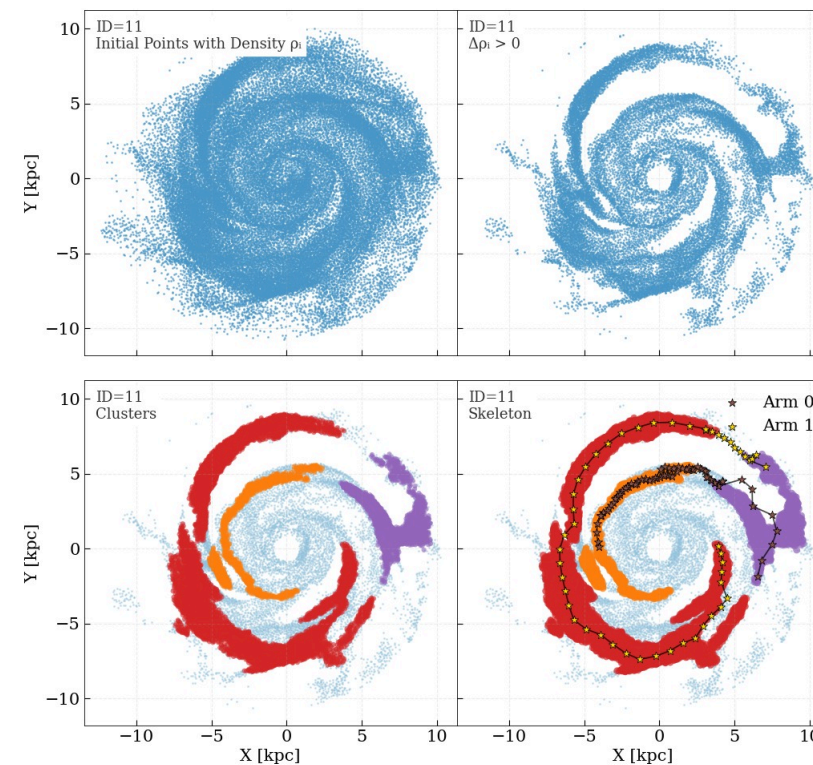
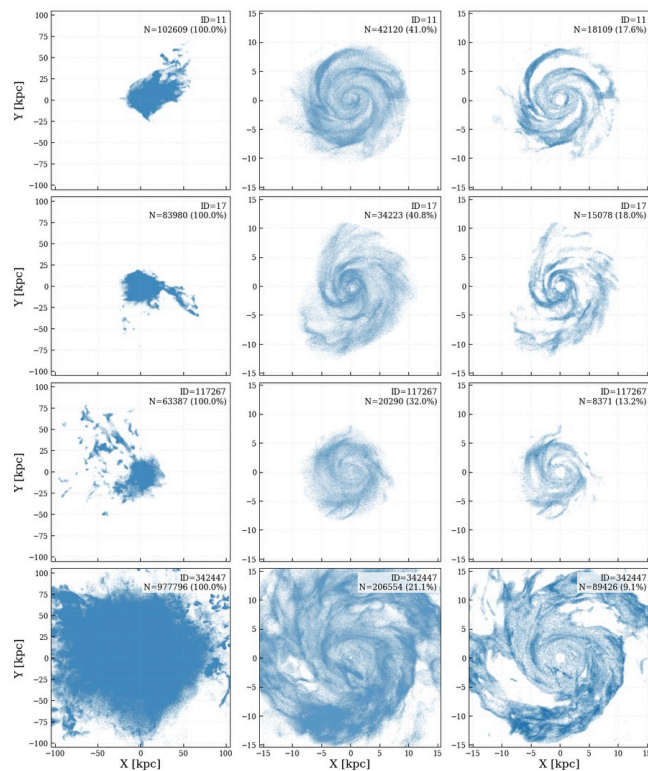
Measuring Morphological Properties

- Select points that trace spiral structure
- Group neighboring points with breadth-first search
- Extract each arm's central path from local centroids
- Estimate pitch angle with a polar-plane fit
- Estimate width and amplitude from transverse density profiles

Exact polar-fit specification remains reserved for the research release.

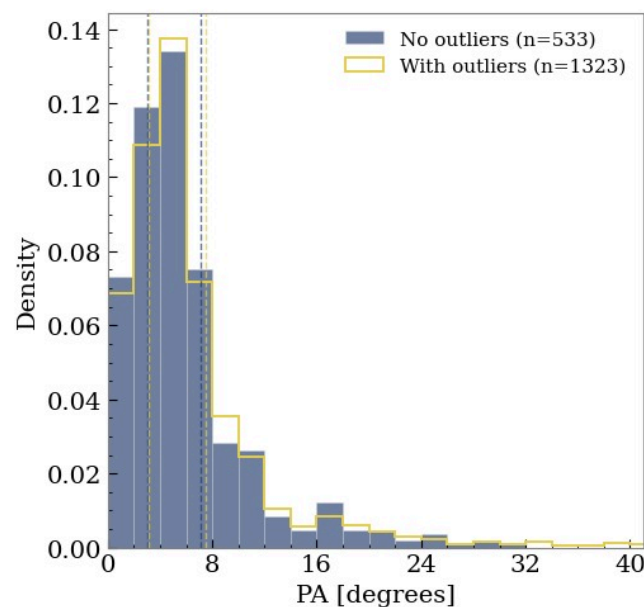


Does the Automated Method Work?

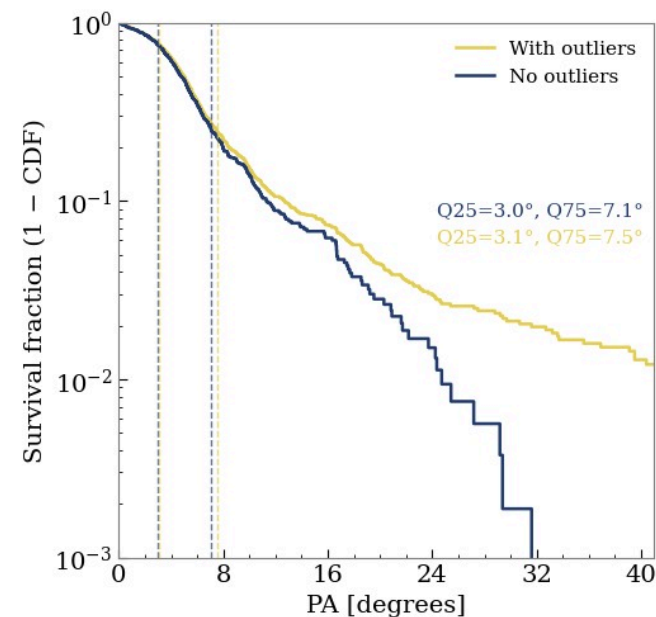


Arms were recovered in about 80% of 309 filtered disks. Failures concentrate in low-density and highly turbulent systems. This is method coverage, not a scientific correlation.

Filtering the Measurement Sample

**1,323**

arm segments in the full catalog

**309**

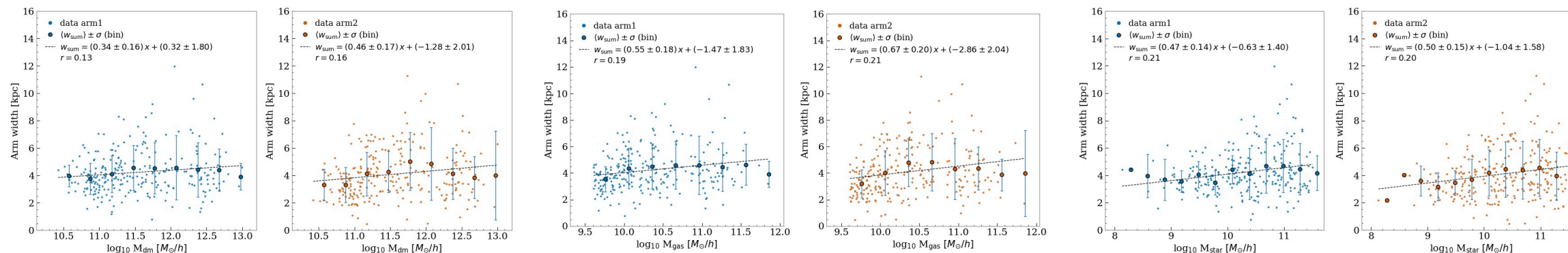
filtered disk galaxies

533

principal arms in a cleaner 267-disk subset

The pitch-angle distribution is strongly concentrated at small absolute angles; the reported mean is 6.85°.

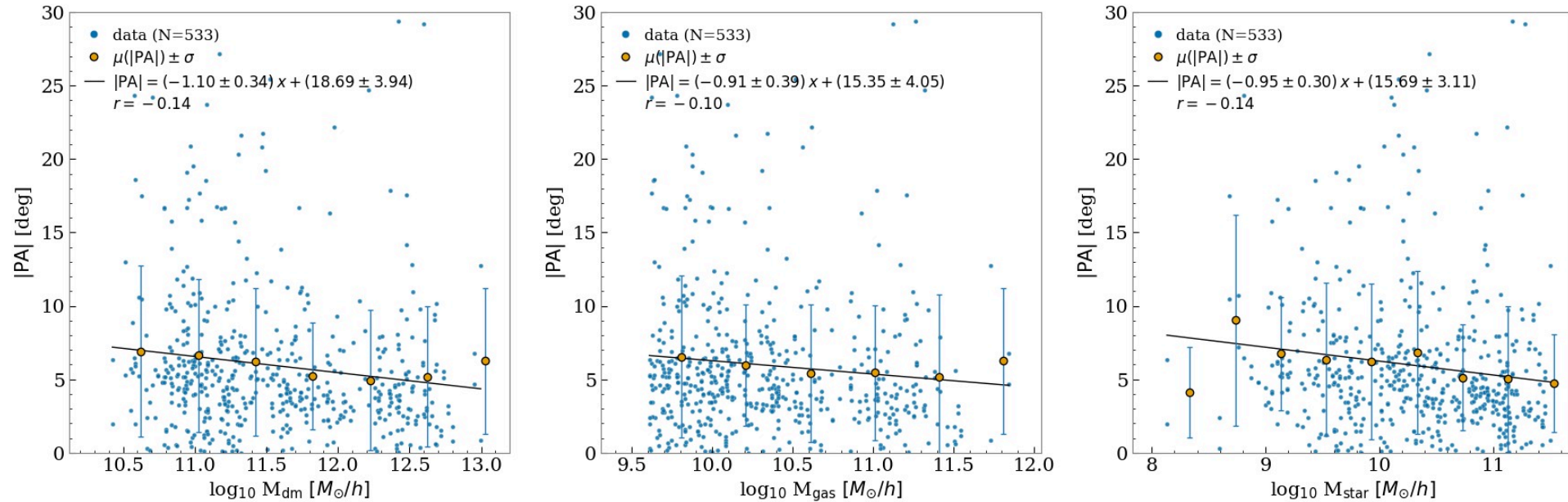
Arm Width vs. Galaxy Mass



Using the original full-disk estimator, mean arm width tends to increase with dark matter, gas, and stellar mass.

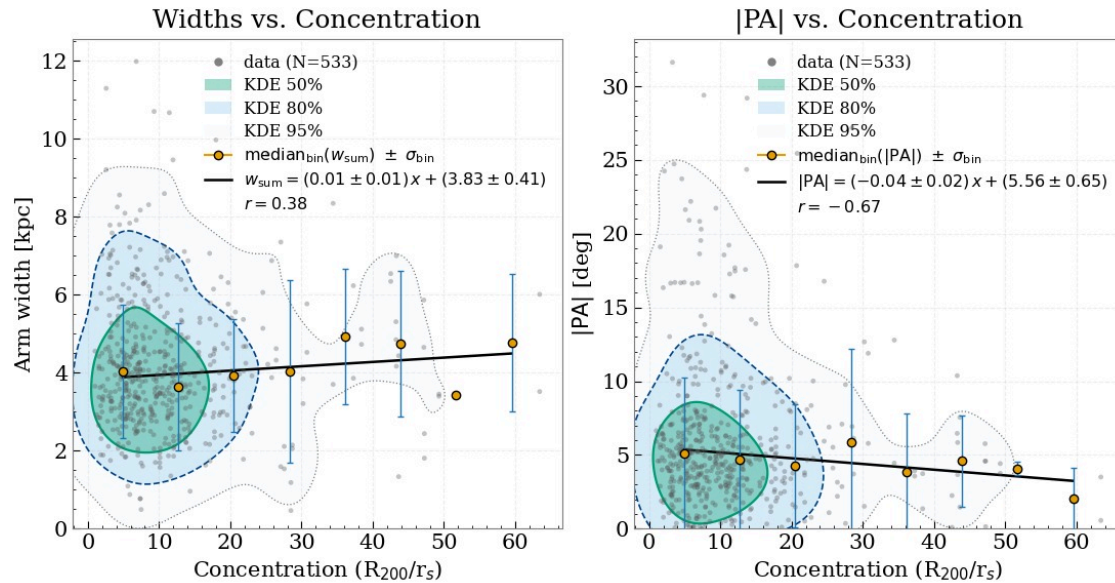
Evidence remains weak: fitted slopes differ from zero by no more than about 1.3σ . A group-only estimator removes a known contamination bias, but has not yet been rerun across the full sample.

Pitch Angle vs. Galaxy Mass



- Absolute pitch angle decreases mildly with mass.
- Differences among dark matter, gas, and stellar mass tracers are not significant.
- More massive systems appear to host slightly more tightly wound arms, but the evidence is not conclusive.

Arm Morphology vs. Halo Concentration



- Mean arm width is consistent with no concentration trend: 0.011 ± 0.011 kpc per unit of c .
- Absolute pitch angle decreases mildly as concentration increases.
- More concentrated halos are associated with slightly tighter winding in this sample.

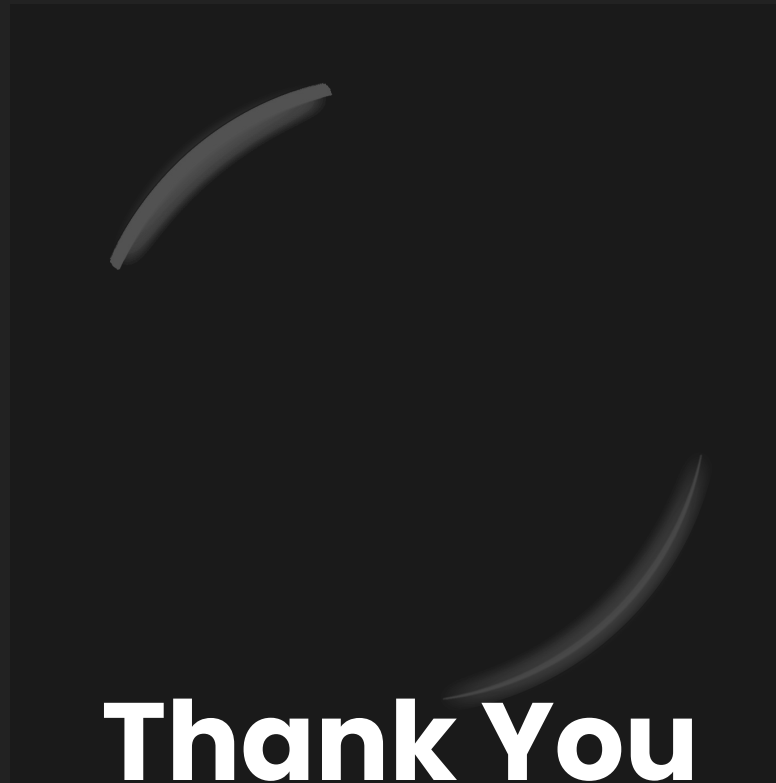
A stronger central potential is a working hypothesis, not a demonstrated causal mechanism.

Conclusions

- We developed an astrophysical and numerical workflow that identifies disk galaxies and extracts coherent spiral-arm structure. Most measurement stages are automated after disk selection and quality control.
- Using the original estimator, mean arm width increases modestly with halo, gas, and stellar mass. This result is tentative and must be retested with the revised width estimator.
- No systematic amplitude–halo–mass trend was detected; reported Pearson coefficients are small ($r \approx 0.06\text{--}0.14$).
- A weak pitch–angle–concentration anti-correlation appears in this sample, but it does not establish a gravitational mechanism.
- Next steps: rerun revised widths across the full sample, enlarge the galaxy sample, quantify selection effects, and compare with observations.

References

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Questions and discussion